Q_8	1	-1	i	j	k
$\#$	1	1	2	2	2
$\mathbb{C}_{++}$	1	1	1	1	1
$\mathbb{C}_{+-}$	1	1	1	-1	-1
$\mathbb{C}_{-+}$	1	1	-1	1	-1
$\mathbb{C}_{--}$	1	1	-1	-1	1
$\mathbb{C}^2$	2	-2	0	0	0

S_4	Id	(12)	(12)(34)	(123)	(1234)
$\#$	1	6	3	8	6
$\mathbb{C}_+$	1	1	1	1	1
$\mathbb{C}_-$	1	-1	1	1	-1
$\mathbb{C}^2$	2	0	2	-1	0
$\mathbb{C}^3_+$	3	-1	-1	0	1
$\mathbb{C}^3_-$	3	1	-1	0	-1

A_5	Id	(123)	(12)(34)	(12345)	(13245)
$\#$	1	20	15	12	12
$\mathbb{C}$	1	1	1	1	1
$\mathbb{C}^3_+$	3	0	-1	$\frac{1+\sqrt{5}}{2}$	$\frac{1-\sqrt{5}}{2}$
$\mathbb{C}^3_-$	3	0	-1	$\frac{1-\sqrt{5}}{2}$	$\frac{1+\sqrt{5}}{2}$
$\mathbb{C}^4$	4	1	0	-1	-1
$\mathbb{C}^5$	5	-1	1	0	0

Indeed, the computation of the characters of the 1-dimensional representations is straightforward.

The character of the 2-dimensional representation of Q_8 is obtained from the explicit formula (4.3.1) for this representation, or by using orthogonality.

For S_4, the 2-dimensional irreducible representation is obtained from the 2-dimensional irreducible representation of S_3 via the surjective homomorphism $S_4 \to S_3$, which allows one to obtain its character from the character table of S_3.

The character of the 3-dimensional representation $\mathbb{C}^3_+$ is computed from its geometric realization by rotations of the cube. Namely, by rotating the cube, S_4 permutes the main diagonals. Thus (12) is